Microproject Guidelines and Implementation

Development of End to End Pipeline for Data Analytics Application Usecase

School of Engineering and Technology

Class: SY AIML AIDS

Dear Students,

All are instructed to prepare and submit the Micro Project as per the following guidelines:

1**. Recent & Relevant Topics:**

Select a topic based on recent trends and practical applications in Data Analytics (preferably related to industry practices, sustainability, or smart technologies).

The problem statements pertaining to following application domains

- Retail and E-commerce
- Healthcare
- Banking and Fintech
- Marketing
- Smart Cities and sustainability

2. **Problem Statement:**

Clearly define the real-world problem you are addressing. The problem should be specific and relevant to AI ML data analytics practical real world industry relevant use cases /applications.

3. Aim:

State one clear and concise aim of the project.

Development of an end-to-end pipeline for data analytics for a specific task/ application

Following listed data analytics pipeline shall be followed

- Problem Definition

- Data Collection
- Data Cleaning
- Exploratory Data Analysis EDA
- Modelling
- Evaluation
- Deployment

4. Objectives:

Mention 3 to 4 measurable objectives that support the achievement of the aim.

- To perform Exploratory Data Analysis (EDA) on specific topic of your choice
- To identify patterns and correlations in data instances
- To apply hypothesis testing for business decision-making
- To build a predictive model for the underlying task chosen

5. Methodology:

Explain step-by-step procedure of how you will carry out the project (data collection, preprocessing, EDA, analysis, modelling, evaluation and deployment etc.) and develop an end-to-end data analytics pipeline for the underlying task.

6. Tools & Software (if any):

Mention use of software and tools frameworks used for data analytics such as python, excel, Power BI, Tableau

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

- Jupyter Notebook

7. Expected Outcomes:

State the possible results or benefits outcomes and societal impact of the project.

The project undertaken shall demonstrates how Data Analytics helps transform raw data into meaningful insights. Statistical inference and predictive modelling support evidence-based decision-making.

8. Conclusion:

Summarize key findings and practical implications.

9. Report Format:

Cover Page

Certificate

Abstract

Index

Introduction

Literature review

Main Contents

Methodology

Results and Discussion

Conclusion and Future Scope

References

📌 The project should be practical-oriented, innovative and neatly presented.

📌 Plagiarism and AI content is strictly not allowed.

Submit the project report on or before the given deadline on 30th March, 2026.

Coverage of topics

1. Dataset collection/ identification and exploration (Primary/ Secondary)

2. Dataset Description
3. Exploratory Data Analysis EDA
4. Stastical Analysis- Mean, Median, Standard Deviation,Variance, Correlation
5. Corelation Analysis
6. Hypothesis Testing H_0: There is no significant relationship
   H_1: There is a significant relationship
7. Predictive Modelling-
8. Linear Regression
9. Logistic Regression
10. Decision Tree

Deployment

Real time dashboards

Github

Presentation-Micro Project Report shall be followed with a presentation on the same with coverage of all above points for assessment with suitable code and deployment.

Dr. Gargi Bhide